Xenon recoil and LHC $t\bar{t}$+MET constraints on an exothermic pseudo-Dirac dark sector

2026-11-10

Abstract

We study a pseudo-Dirac dark-matter model in which a vector mediator V couples to quarks with a mass-weighted strength, $g_{Vq} = g_t\,\hat{m}_q/172$, and to the two nearly degenerate dark states through an off-diagonal current. The model has four free parameters: the dark-matter mass m_χ, the mediator mass M_V, and the two couplings g_t and g_χ; the mass splitting is fixed at $\delta \equiv m_{\chi_2} - m_{\chi_1} = 1$ MeV. We impose three constraints simultaneously: the xenon recoil rate reported by LZ, the thermal relic abundance measured by Planck ($\Omega_\chi h^2 = 0.12$), and the CMS search for $t\bar{t}$+MET at $\sqrt{s} = 13$ TeV with 138 fb^{-1}. The first two reduce the parameter space to the (M_V, g_t) plane, on which the third defines a 95% CL exclusion region. At $M_V = 20$ GeV the whole column survives; for $M_V = 30$–50 GeV the allowed region splits into two g_t windows with an excluded band in between; for $M_V = 60$–80 GeV only a narrow strip $g_t \lesssim 0.35$–0.38 remains; above $M_V \simeq 90$ GeV the entire valid parameter range is excluded. The two windows arise because a small g_t suppresses the signal as g_t^2, while a large g_t drives $\Gamma(V \to q\bar{q}) \propto g_t^2$ and crushes BR$(V \to \chi\chi)$.

Contents

1. Introduction **2**
2. Model **2**
3. Direct detection: LZ rate normalisation **3**
4. Thermal relic abundance **4**
5. LHC $t\bar{t}$+MET analysis **5**
   - 5.1 Signal and selection . . . 5
   - 5.2 Statistical model . . . 6
   - 5.3 Independent validation of the statistical framework . . . 6
   - 5.4 Limits . . . 7
6. Combined constraints and exclusion region **8**
   - 6.1 The (M_V, g_t) plane . . . 8
   - 6.2 Origin of the two g_t windows . . . 8
   - 6.3 Results . . . 9
   - 6.4 Systematic uncertainties . . . 10
7. Conclusions **11**

References **12**

1. Introduction

If dark matter scatters off nuclei exothermically — that is, if the incoming state is the heavier of the two mass eigenstates — then even very slow dark matter can produce nuclear recoils of several tens of keV. LZ has reported a candidate event at 248 keV in this energy range with a local significance of about 2.6σ. A single event neither establishes a discovery nor excludes a model, but it provides a quantitative constraint: if the event is due to dark-matter scattering, the scattering rate at the corresponding dark-matter mass must reproduce exactly the observed number of events.

We combine this condition with two independent constraints:

- **Direct detection:** the LZ recoil rate depends only on the combination $\kappa = g_t g_\chi / M_V^2$, and therefore fixes a surface in κ;
- **Thermal relic abundance:** the Planck value of $\Omega_\chi h^2$ fixes m_χ for each M_V;
- **Collider:** the CMS $t\bar{t}$+MET search sets an upper limit on g_t, which feeds back on g_t itself through $\mathrm{BR}(V \to \chi\chi)$.

Section 2 introduces the model; Sections 3 and 4 give the normalisation of the first two constraints; Section 5 presents the $t\bar{t}$+MET analysis; Section 6 gives the combined exclusion region; Section 7 summarises.

2. Model

The dark sector consists of two nearly degenerate Majorana states and a vector mediator,

$$\chi_1,\ \chi_2 \quad (m_{\chi_2} = m_{\chi_1} + \delta), \qquad V_\mu\ (M_V), \qquad \delta = 1\ \text{MeV}.$$

The Lagrangian is

$$\mathcal{L} \supset -\tfrac{1}{4} V_{\mu\nu} V^{\mu\nu} + \tfrac{1}{2} M_V^2 V_\mu V^\mu + \tfrac{1}{2}\, V_\mu\, \bar{\chi}_2\, (g_\chi \gamma^\mu)\, \chi_1 + \text{h.c.} - V_\mu \sum_q g_{Vq}\, \bar{q} \gamma^\mu q, \quad (1)$$

$$g_{Vq} = g_t\, \frac{\hat{m}_q}{172\ \text{GeV}}. \quad (2)$$

By Eq. (2) the coupling is linear in the quark mass and therefore largest for the top quark ($g_{Vt} = g_t$), which is what makes the $t\bar{t}$+MET channel sensitive. The relic abundance is solved on the grid $5 \le m_\chi \le 56$ GeV, $20 \le M_V \le 200$ GeV; the collider scan covers

$$20 \le M_V \le 200\ \text{GeV}, \qquad 0.1 \le g_t \le \sqrt{4\pi} = 3.545,$$

and we require $\Gamma_V / M_V < 3\%$ (narrow width) and $M_V - 2m_\chi \ge 5\Gamma_V$ (kinematically open).

The model is implemented as a UFO. The signal samples were generated with `PseudoDiracFlavorEFT_UFO_loop`, in which the six quark couplings are independent external parameters (`Block flavcoup` entries 1–6) that the parameter card sets one by one according to Eq. (2); for $g_t = 1$ they are 1.279×10^{-5}, 2.733×10^{-5}, 5.407×10^{-4}, 7.384×10^{-3}, 2.430×10^{-2} and 1.000. The g_χ and δ values go into `PDBINPUTS`, the mediator and dark-matter masses into `Block mass` (`9900032`, `100051`, `100052`). A second, single-parameter packaging, `PseudoDiracFlavorEFT_UFO_massprop` ($c_V = g_t/172$ in `PDBINPUTS`, $g_{Vq} = c_V \hat{m}_q$), has identical Feynman rules and gives 5.905 versus 5.836 pb at $M_V = 100$ GeV, $g_t = 1$ (a 1.17% difference); it serves as a cross-check of the coupling conventions. Signal samples are generated with MadGraph5_aMC@NLO 3.6.6, hadronised with Pythia8, passed through Delphes, and subjected to the analysis selection.

3. Direct detection: LZ rate normalisation

For exothermic scattering $\chi_2 N \to \chi_1 N$ the minimum incoming speed is

$$v_{\min} = \left| \frac{q}{2\mu_N} - \frac{\delta}{q} \right|, \qquad \mu_N = \frac{m_\chi m_N}{m_\chi + m_N}, \qquad q = \sqrt{2 m_N E_R}. \tag{3}$$

In Eq. (3) the δ term is negative, which pushes $v_{\min}$ well below its elastic value; this is what makes a 248 keV recoil accessible for $m_\chi \sim 10$ GeV.

The rate factorises in the model parameters, $R \propto (g_t g_\chi)^2 / M_V^4 \equiv \kappa^2$, so requiring one event in the region of interest fixes a single combination:

$$\kappa_{\rm LZ}(m_\chi) = \frac{g_t g_\chi}{M_V^2}, \qquad \Longrightarrow \qquad g_\chi = \frac{\kappa_{\rm LZ}(m_\chi)\, M_V^2}{g_t}. \tag{4}$$

Equation (4) is the first constraint and removes one degree of freedom. The calculation uses the coherent recoil spectrum of nine xenon isotopes ($\rho_\chi = 0.3$ GeV/cm^3, $v_0 = 238$, $v_E = 254$, $v_{\rm esc} = 544$ km/s, ROI 5.4–269.9 keV, exposure 2.84 t·yr) with an excited-state fraction $f_{\rm exc} = 0.5$. The quantities g_t and g_χ in Eq. (4) are model parameters, whereas the recoil code needs a nucleon coupling. For the vector current with $g_{Vq} = g_t \hat{m}_q / 172$ the nucleon matrix element at $q^2 \to 0$ is the valence sum

$$g_N(p) = 2 g_{Vu} + g_{Vd} = g_t \times 5.291 \times 10^{-5},$$

i.e. 5.291×10^{-5} at $g_t = 1$; the same model file (`vars1.mdl`) quotes $g_{Vu} = 1.279 \times 10^{-5}$ and $g_{Vd} = 2.733 \times 10^{-5}$, in exact agreement. The rate is proportional to $(g_N g_\chi)^2 / M_V^4$; the normalisation constant $1/16\pi$ in the code corresponds to the $1/2$ on the dark vertex in Eq. (1).

Solving point by point on 36 mass values (choosing g_χ such that the region of interest contains exactly one event; the 9–26 GeV range is sampled every 0.5 GeV) gives the curve of Fig. 1(b); the joint constraint is evaluated at the 14 abundance solutions:

$$\kappa(8.53) = 2.68 \times 10^{-5}, \qquad \kappa(12.03) = 9.37 \times 10^{-5},$$

$$\kappa(24.10) = 5.73 \times 10^{-5}, \qquad \kappa(37.86) = 1.10 \times 10^{-4}\ \text{GeV}^{-2}.$$

The curve is not monotonic: it peaks at $m_\chi \approx 13$ GeV (1.17×10^{-4}), falls back to 5.6×10^{-5} at $m_\chi \approx 22$ GeV, and then rises slowly. The peak and the dip come from exothermic kinematics: which part of the region-of-interest window the recoil spectrum falls into varies steeply with m_χ, and near $m_\chi \approx 13$ GeV the minimum of $v_{\min}$ matches the ROI threshold worst, so the same event rate requires a larger coupling. This is also why a few-point interpolation cannot replace the point-by-point solution.

κ scales as $f_{\text{exc}}^{-1/2}$; with $f_{\text{exc}} = 0.1$ it increases by a factor 2.19, which is the largest single systematic uncertainty on the direct-detection axis.

The meaning of κ changes with the dark-matter mass. A 248 keV true recoil has a kinematic floor: from Eq. (3) with $q = \sqrt{2m_N E_R} = 0.246$ GeV the minimum speed $v_{\min} = |q/(2\mu_N) - \delta/q|$ is 3400, 2150, 1380 and 910 km/s at $m_\chi = 8.5$, 12, 16 and 20 GeV, all above $v_{\text{esc}} + v_E = 798$ km/s; the lightest mass that can produce a 248 keV recoil is about 21.5 GeV. Hence for the $M_V \leq 50$ GeV columns ($m_\chi = 8.5$–20.2 GeV) κ means "one event in total in the region of interest", not "one event at 248 keV": the predicted recoil spectrum there is softer, 5–100 keV. Only the $M_V \geq 60$ GeV columns ($m_\chi \geq 24.1$ GeV) can also match the energy.

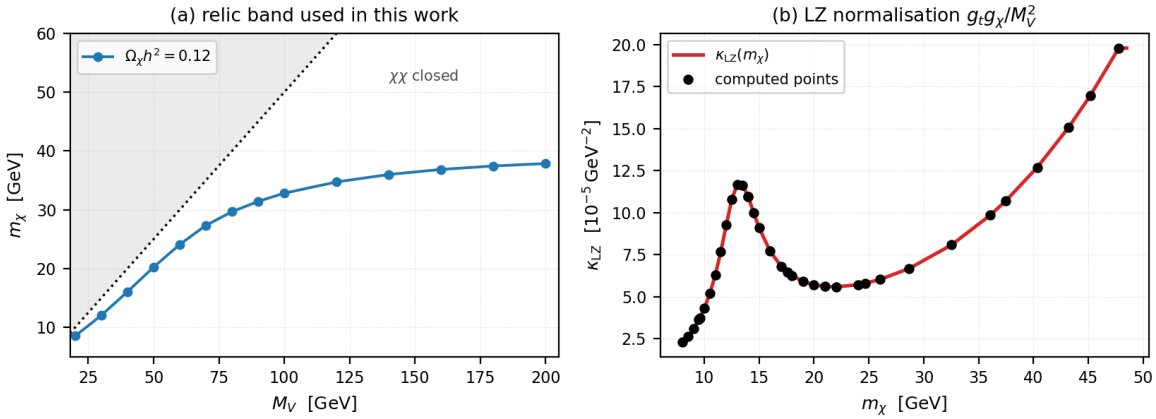


Figure 1: Normalisation of the LZ and Planck constraints. (a) $m_\chi(M_V)$ obtained from $\Omega_\chi h^2 = 0.12$, 14 mass points ($m_\chi = 8.53$–37.86 GeV); the dotted line is $m_\chi = M_V/2$ and the grey region above it is where $\chi\chi$ is kinematically closed. (b) $\kappa_{\text{LZ}}(m_\chi)$ (red line) with 36 computed points (black dots; the 9–26 GeV range is sampled every 0.5 GeV).

4. Thermal relic abundance

The two states are almost degenerate, so before freeze-out $\chi_1\chi_1$, $\chi_1\chi_2$ and $\chi_2\chi_2$ annihilate simultaneously, with the effective cross section

$$\langle\sigma v\rangle_{\rm eff} = \tfrac{1}{4}\langle\sigma v\rangle_{11} + \tfrac{1}{2}\langle\sigma v\rangle_{12} + \tfrac{1}{4}\langle\sigma v\rangle_{22}, \tag{5}$$

and the abundance follows from the standard freeze-out equation. Neglecting co-annihilation (keeping only the first term of Eq. 5) underestimates Ωh^2 by about a factor of two, so it must be included.

We solve `darkOmega` with micrOMEGAs 7.1 on a (m_χ, M_V) grid: m_χ from 5 to 56 GeV and M_V from 20 to 200 GeV, 20–26 points per column, refined to 0.3 GeV around the $\chi\chi$ threshold $m_\chi = M_V/2$. At every grid point g_χ is fixed by the LZ condition of Eq. (4) ($g_t = 1$, $g_\chi = \kappa_{\rm LZ} M_V^2$), and we then impose the Planck band $0.1176 \le \Omega_\chi h^2 \le 0.1224$. At the reference point $m_\chi = 35$ GeV, $M_V = 87.68$ GeV, $g_\chi = 0.180$, $g_t = 1.80$ we reproduce $\Omega h^2 = 0.123$, 2.5% from the target, which validates the chain. Since $\mathrm{d}\Omega/\mathrm{d}m_\chi$ is steep, the $\pm1\sigma$ abundance band corresponds to an O(1%) change in m_χ, so each M_V has a single solution at the centre of the band, $m_\chi^{\rm relic}(M_V)$. Figure 1(a) shows this band: m_χ rises monotonically from 8.53 GeV at $M_V = 20$ GeV to 37.86 GeV at $M_V = 200$ GeV, always satisfying $2m_\chi < M_V$, i.e. the $\chi\chi$ channel is open along the whole band.

The $M_V = 20$ GeV column deserves a comment: at the $\chi\chi$ threshold $m_\chi = M_V/2 = 10$ GeV the resonance drives Ωh^2 down to 1.3×10^{-4}, and $\Omega h^2 = 0.12$ has a solution on each side (8.53 and 10.34 GeV). We take 8.53 GeV, which satisfies $2m_\chi < M_V$; the other solution lies above threshold, where the $\chi\chi$ channel is closed and contributes nothing to the $t\bar{t}$+MET signal. At this solution $2m_\chi/M_V = 0.85$, 2.9 GeV below threshold, so it is no longer on the steep resonance flank; with the 0.3 GeV grid the root is obtained by interpolation between adjacent points and is accurate to about 0.05 GeV (below 2% in κ). The other 13 solutions all lie in $0.38 \le 2m_\chi/M_V \le 0.81$: for $M_V \le 60$ GeV they sit about 20% below the s-channel V pole (the resonance funnel) and move away as M_V grows, reaching 0.38 at 200 GeV. No column lies on the steep flank at $2m_\chi = M_V$.

5. LHC $t\bar{t}$+MET analysis

5.1 Signal and selection

The signal is $pp \to t\bar{t}\,V$ with $V \to \chi_1\chi_2$, giving a top-quark pair plus large $p_T^{\rm miss}$. We use the AH $0\ell2b$ region of CMS EXO-22-014: zero leptons, $n_{\rm jet} \ge 3$ ($p_T > 30$ GeV, $|\eta| < 2.4$), $n_b \ge 2$, $p_T^{\rm miss} > 250$ GeV, $\min\Delta\phi(j_{1,2}, p_T^{\rm miss}) > 0.8$, $m_T^b > 140$ GeV and $p_T(j_1)/H_T < 0.5$, for an integrated luminosity of 138 fb^{-1}.

The signal yield is

$$s_i(M_V, g_t) = \varepsilon_i(M_V)\;\sigma_{t\bar{t}V}(M_V)\;g_t^2\;\mathrm{BR}(V \to \chi\chi; g_\chi, g_t)\;\mathcal{L}, \tag{6}$$

where ε_i is the efficiency and shape in the 15 $p_T^{\rm miss}$ bins, measured on 2.6–2.7×10^4 full-chain events per mass point (164–1346 passing, efficiency 0.61%–4.92%).

5.2 Statistical model

The statistical inputs are taken directly from the CMS post-fit results: signal, total background, background components and observed counts in 15 bins, with the per-bin background uncertainty reconstructed from the quadrature sum of the components (Fig. 2). The likelihood is

$$-2\ln L(\mu,\theta) = \sum_{i=1}^{15}\Big[-2\big(n_i\ln\big(b_i(1+\theta)+\mu s_i\big)-\big(b_i(1+\theta)+\mu s_i\big)\big)\Big] + \left(\frac{\theta}{w}\right)^2, \qquad w=\frac{\sum_i \sigma_i}{\sum_i b_i}=3.26\%, \tag{7}$$

i.e. one background nuisance θ per signal region (coherent within the region; the constraint term is the second line of Eq. 7). The test statistic is $\tilde{q}_\mu$, CLs is computed with the asymptotic formulae of Cowan et al., and the expected limit uses the Asimov data set.

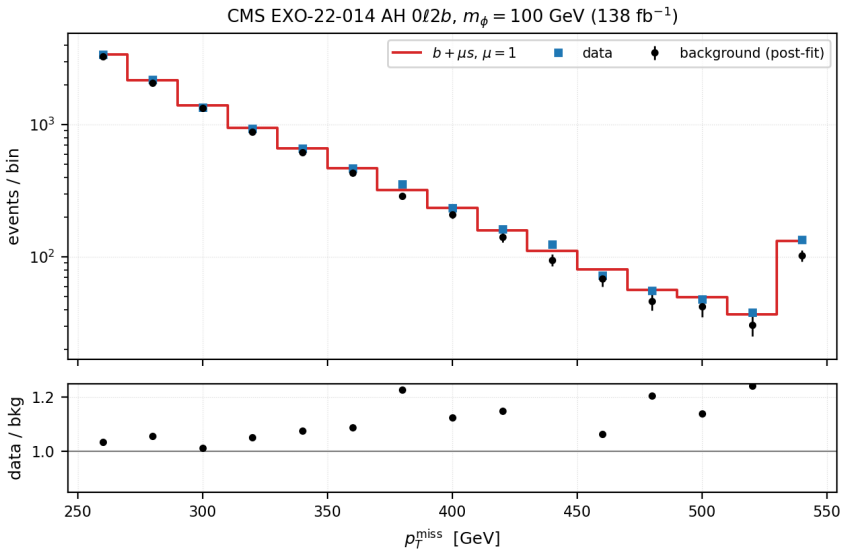


Figure 2: Statistical inputs of the CMS EXO-22-014 AH $0\ell 2b$ region ($m_\phi = 100$ GeV, 138 fb^{-1}): total background with per-bin uncertainties (black), background plus reference signal (red) and observed data (blue). Lower panel: data-to-background ratio.

5.3 Independent validation of the statistical framework

The statistical chain is cross-checked at two levels (the itemised records are in `cmsrepro/phase2/PHASE_2_REPORT.md` and `cmsrepro/phase25/PHASE_25_REPORT.md`).

Single signal region. The likelihood of Eq. (7) is compared point by point with the HEP-standard implementation pyhf 0.7.6 on the same inputs (Table 1):

Table 1: Comparison of the likelihood of Eq. (7) with pyhf 0.7.6 on the same inputs (AH $0\ell 2b$, single region).

	$\hat{\mu}$	μ_{95} (this work)	μ_{95} (pyhf)	deviation
Expected (Asimov)	0.000	0.3959	0.3963	−0.10%
Observed	1.0465	1.3893	1.3896	−0.02%

All signal regions. The CMS record contains 11 signal regions (AH 3, SL 6, DL 2) with 99 bins in total. Combining them in a joint likelihood with a shared μ, the agreement with pyhf is 0.04% (expected) and 0.01% (observed).

Signal normalisation. The production cards published by CMS show that the signal is generated from the 0-parton and 1-parton sub-processes of $pp \to t\bar{t}V$ with MLM matching and a fixed scale of 91.188 GeV, so the signal templates correspond to the combined LO cross section $\sigma_{\text{ref}} = 1.06241$ pb. The ratio of this value to the 0-parton table rises monotonically from 1.29 at 50 GeV to 1.95 at 200 GeV, consistent with one additional parton.

5.4 Limits

Table 2 collects the 95% CL limits. μ_{95} is the dimensionless signal strength relative to the CMS template; the cross-section limits in the third and fifth columns follow from $\sigma_{95} = \mu_{95}\,\sigma_{\text{ref}}$ with $\sigma_{\text{ref}} = 1.06241$ pb. Row 1 is the single region, row 2 the 11-region combination with per-bin independent errors and row 3 the same with the coherent errors of Eq. (7). All calculations use the CMS signal templates, so they can be compared directly with the official row.

Table 2: Summary of the 95% CL limits ($\sigma_{\text{ref}} = 1.06241$ pb, $\sigma_{95} = \mu_{95}\sigma_{\text{ref}}$).

calculation	μ_{95} exp	σ_{95} exp [pb]	μ_{95} obs	σ_{95} obs [pb]
single region $0\ell 2b$	0.396	0.421	1.389	1.476
11 regions (independent)	0.232	0.246	0.954	1.013
11 regions (coherent)	0.258	0.274	0.784	0.833
CMS official	—	0.295	—	0.532

The combination is about 35% stronger than the single region, consistent with AH $0\ell 2b$ dominating the sensitivity. The residual difference from the official value has two sources: the official fit includes correlated systematic uncertainties across regions and the control-region constraints, whereas our inputs are the post-fit per-bin errors.

One further caveat applies to the observed column: the background central values of CMS Fig. 2 come from a joint signal-plus-background fit to the same data. Treating the post-fit background as an independent prior and then applying the ob-

served counts of the same signal region re-uses the data, so the observed limits are indicative only; the validation of the statistical framework rests on the agreement of the expected column with pyhf (0.04%).

6. Combined constraints and exclusion region

6.1 The (M_V, g_t) plane

Inserting Eq. (4) and the abundance solution of Section 4 reduces the parameter space to two dimensions:

$$g_\chi = \frac{\kappa_{\rm LZ}(m_\chi) M_V^2}{g_t}, \qquad m_\chi = m_\chi^{\rm relic}(M_V). \tag{8}$$

Each column M_V then fixes m_χ, $g_\chi(g_t)$, $\Gamma_V(g_t)$ and $\mathrm{BR}(V \to \chi\chi)$; where several abundance solutions exist for one M_V we keep the branch with $2m_\chi < M_V$ (only $M_V = 20$ GeV has two, and we take 8.53 GeV). Substituting Eq. (8) into Eq. (6) gives the signal yield of that column.

6.2 Origin of the two g_t windows

The contour structure of Fig. 3 is governed by two opposing scalings:

$$s \propto g_t^2\ \mathrm{BR}(V \to \chi\chi), \qquad \mathrm{BR} = \frac{\Gamma_{\chi\chi}}{\Gamma_{\chi\chi} + \Gamma_{q\bar{q}}}, \qquad \Gamma_{q\bar{q}} \propto g_t^2 \sum_q \left(\frac{m_q}{172}\right)^2. \tag{9}$$

By Eq. (9), a small g_t suppresses the yield through g_t^2 and the signal falls below the background; a large g_t makes $\Gamma_{q\bar{q}}$ grow as g_t^2 and crushes BR, so the signal vanishes again. In between, μ_{95} has a minimum and defines an excluded interval. For $M_V = 30$–50 GeV this interval does not reach $g_t = 0$ or $g_t = \sqrt{4\pi}$, so the allowed region appears as two disconnected g_t windows; above $M_V \simeq 60$ GeV the large-g_t side is excluded entirely and only a narrow low-g_t strip survives. Figure 4 shows $\mu_{95}(g_t)$; the window edges are its intersections with $\mu_{95} = 1$.

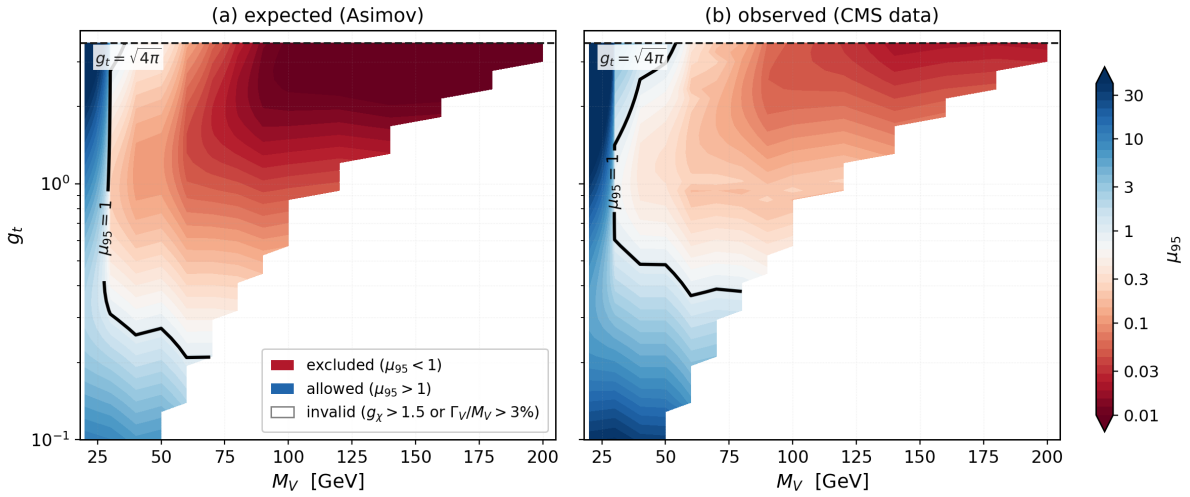


Figure 3: 95% CL exclusion in the (M_V, g_t) plane. (a) expected (Asimov); (b) observed (CMS data). Dark red is excluded ($\mu_{95} < 1$), dark blue allowed, and white marks invalid parameters ($g_\chi > 1.5$ or $\Gamma_V/M_V > 3\%$). The black line is $\mu_{95} = 1$, the dashed line at the top is $g_t = \sqrt{4\pi}$. The figure uses the full-chain efficiency of this work (the ε_i of Eq. 6) and is therefore not the same quantity as the CMS-template limits of Table 2.

6.3 Results

Table 3: Minimum μ_{95} per M_V column and the allowed g_t interval (observed, with the error model of Table 2).

M_V [GeV]	min μ_{95} (exp)	min μ_{95} (obs)	verdict
20	2.07	5.71	column allowed
30	0.221	0.720	$g_t \leq 0.57$ or ≥ 1.42
40	0.105	0.366	$g_t \leq 0.48$ or ≥ 2.76
50	0.103	0.317	$g_t \leq 0.45$ or ≥ 3.00
60	0.042	0.177	$g_t \lesssim 0.35$
70	0.028	0.146	$0.21 \leq g_t \leq 0.38$
80	0.018	0.096	$0.32 \leq g_t \leq 0.38$
90	0.011	0.053	column excluded
100	8.6×10^{-3}	0.046	column excluded
120	6.0×10^{-3}	0.033	column excluded
140	4.2×10^{-3}	0.020	column excluded
160	4.0×10^{-3}	0.021	column excluded
180	4.2×10^{-3}	0.022	column excluded ($g_t \geq 2.34$)
200	3.9×10^{-3}	0.020	column excluded ($g_t \geq 3.00$)

The large white area in the lower-right of Fig. 3 is produced by the validity requirement $g_\chi \leq 1.5$: by Eq. (8), $g_\chi = \kappa M_V^2/g_t$ grows as M_V^2 at fixed g_t, so every column requires $g_t \geq \kappa M_V^2/1.5$ — 0.14 from $M_V = 60$ GeV, 0.55 from 100 GeV,

1.28 from 140 GeV, 2.31 from 180 GeV and 2.93 from 200 GeV. The last value is still below the perturbativity bound $\sqrt{4\pi} = 3.545$, so the 200 GeV column retains a narrow valid strip at the top. The bound $\Gamma_V/M_V \leq 3\%$ only acts at smaller g_t, with a boundary to the left of $g_\chi = 1.5$, so the white edge seen in the figure is set by the g_χ criterion.

For $M_V = 30$–50 GeV both ends of the two windows lie inside $g_\chi < 1.5$ and $\Gamma_V/M_V < 3\%$ (the tightest point is $M_V = 50$ GeV, $g_t = 0.1$, with $g_\chi = 1.42$), so these windows are genuine μ_{95} crossings rather than artefacts of the validity cuts.

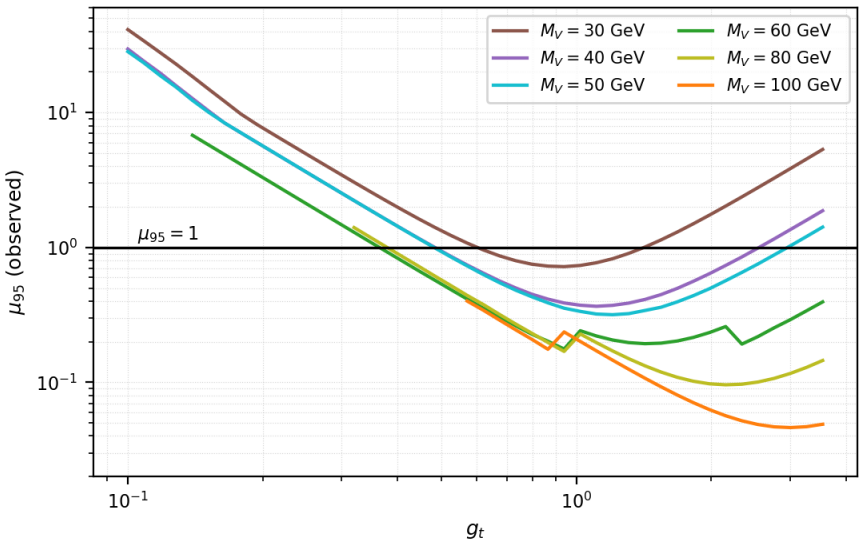


Figure 4: μ_{95} versus g_t at fixed M_V (observed). The curves fall and then rise; their intersections with $\mu_{95} = 1$ are the window edges of Fig. 3, and the $M_V = 30$, 40 and 50 GeV curves have two intersections each.

6.4 Systematic uncertainties

Table 4 summarises the systematic uncertainties and the direction in which they move the exclusion region.

Table 4: Systematic uncertainties and their effect on the exclusion region.

source	effect
f_{exc} (0.5 versus 0.1)	κ changes by a factor 2.19, the allowed region shrinks
single region, no control regions	observed limit about 50% weaker
coherent errors instead of the official covariance	expected limit 7%–16% stronger
full-chain efficiency not aligned with the CMS cut-flow	exclusion region systematically larger
no NLO corrections	signal rate too low, g_t limit conservative

source	effect
nucleon normalisation of κ_{LZ} (valence-quark prescription for g_N)	shifts the windows overall; the equal-weight $g_B/6$ prescription gives a g_N that differs by 7.25
signal templates generated at the previous band's m_χ (3%–26% higher, above 14% for $M_V \geq 100$)	samples not regenerated; about 10% extra uncertainty on the $M_V = 70$–80 GeV window edges

The value of κ_{LZ} depends on the nucleon-coupling convention. We use the mass-weighted couplings of the model itself, $g_N(p) = 2g_{Vu} + g_{Vd} = 5.291 \times 10^{-5}$ at $g_t = 1$, which are the same parameters as $g_{Vq} = g_t \widehat{m}_q/172$, so there is no additional normalisation uncertainty. In the alternative convention often used in the literature — g_B shared equally among six quarks, $g_{Vq} = g_B/6$ — the valence sum gives $g_B/2$, a factor 7.25 smaller than ours, and the corresponding κ differs by the same factor, translating the exclusion region as a whole. We do not adopt that convention: its quark couplings are flavour-blind, so it is not the same Lagrangian as the model studied here (whose top coupling is the largest), and the two g_t cannot be converted into one another by a single number. Table 3 and Fig. 3 use our convention throughout.

7. Conclusions

We have combined the LZ xenon-recoil candidate, the Planck thermal relic abundance and the CMS $t\bar{t}$+MET search within a single parameterisation. The conclusions are:

1. The first two constraints reduce the four-dimensional parameter space to the (M_V, g_t) plane: g_χ follows from Eq. (4) and m_χ from $\Omega_\chi h^2 = 0.12$.
2. At $M_V = 20$ GeV the whole column survives; for $M_V = 60$–80 GeV only a narrow strip $g_t \lesssim 0.35$–0.38 remains (the lower edge at $M_V = 70$ and 80 GeV is set by $g_\chi \leq 1.5$, at 0.21 and 0.32); above $M_V \simeq 90$ GeV the entire valid parameter range is excluded.
3. For $M_V = 30$–50 GeV two g_t windows appear (e.g. $g_t \leq 0.48$ and $g_t \geq 2.76$ at $M_V = 40$ GeV). They follow from the opposing scalings of g_t^2 and $\mathrm{BR}(V \to \chi\chi)$ in Eq. (9) and are not statistical fluctuations; both windows lie inside $g_\chi < 1.5$ and $\Gamma_V/M_V < 3\%$.
4. The statistical framework is independently validated against pyhf (0.10%/0.02% for the single region, 0.04%/0.01% for the 11-region combination) and the signal normalisation is fixed by the CMS production cards at 1.06241 pb, so Table 2 can be compared directly with the official limits.

Further work includes control-region constraints to recover the observed sensitivity, the official covariance matrix, NLO corrections to the signal, and a bin-by-bin alignment of the efficiency curve with the CMS cut-flow.

References

[1] CMS Collaboration, *Search for dark matter produced in association with top quarks in proton–proton collisions at* $\sqrt{s} = 13$ *TeV*, JHEP 08 (2025) 085, arXiv:2505.05300.

[2] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, A&A 641 (2020) A6.

[3] G. Cowan, K. Cranmer, E. Gross, O. Vitells, *Asymptotic formulae for likelihood-based tests of new physics*, Eur. Phys. J. C 71 (2011) 1554.

[4] L. Heinrich et al., *pyhf: pure-Python implementation of HistFactory statistical models*, JOSS 6 (2021) 3173.

[5] G. Bélanger, F. Boudjema, A. Pukhov, A. Semenov, *micrOMEGAs: a tool for dark matter studies*, Comput. Phys. Commun. 192 (2015) 322.

[6] J. Alwall et al., *The automated computation of tree-level and next-to-leading order differential cross sections*, JHEP 07 (2014) 079.

[7] T. Sjöstrand et al., *An introduction to PYTHIA 8.2*, Comput. Phys. Commun. 191 (2015) 159.

[8] J. de Favereau et al., *DELPHES 3: a modular framework for fast simulation of a generic collider experiment*, JHEP 02 (2014) 057.